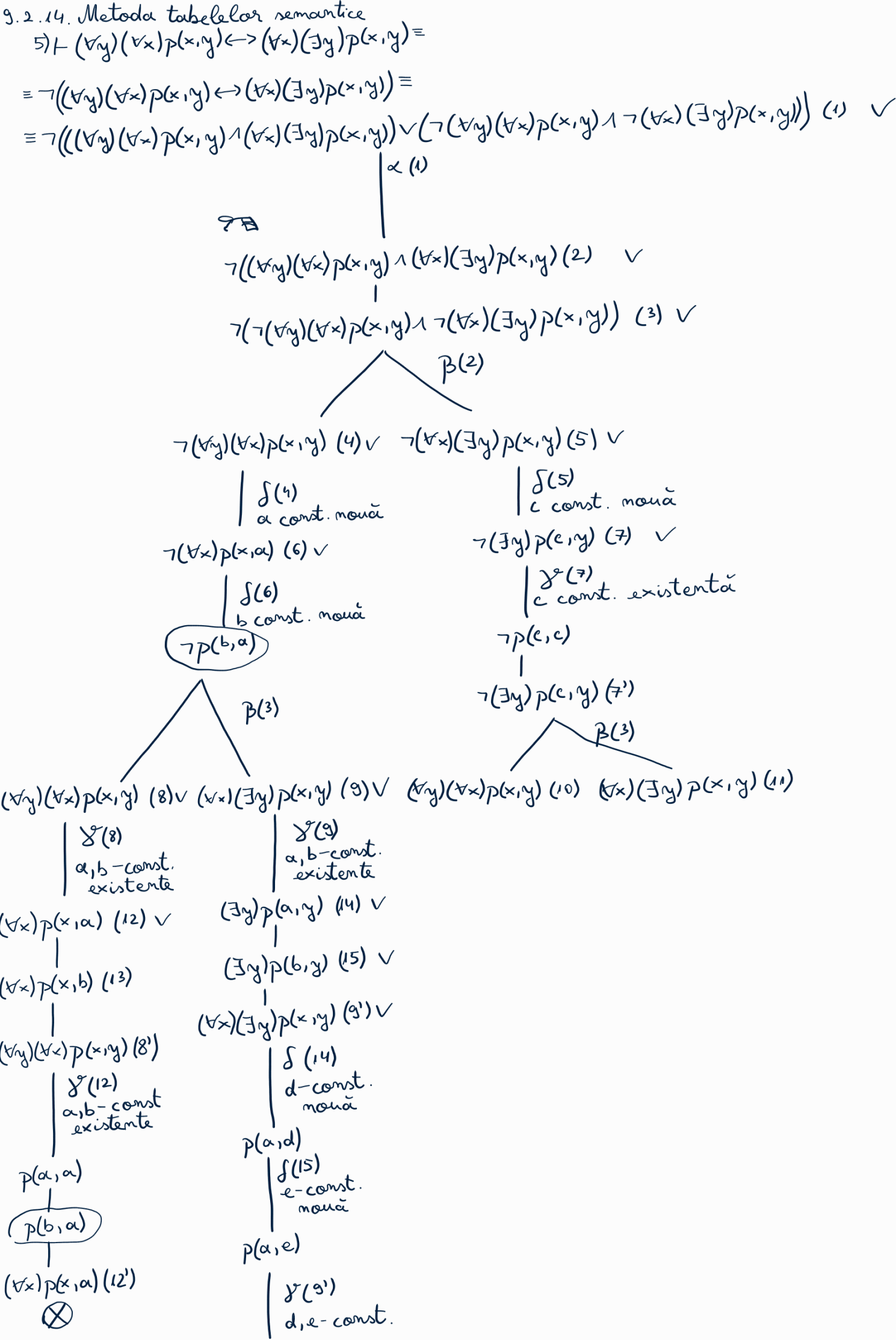




| existente  
 $(\exists y)P(a, y) \text{ (16)}$   
 |  
 $(\exists y)P(e, y) \text{ (17)}$   
 |  
 $(\forall x)(\exists y)P(x, y) \text{ (9'')}$   
 ciclu infinit  $\Rightarrow$  nu putem decide tipul formulei

g. 2.8.5. Sunt unificabili atomii din perechile următoare? Dacă da, identificați c.m. general unificator?

$$\begin{aligned} A_1 &= P(a, x, g(f(y))) \\ A_2 &= P(f(z), z, g(x)) \\ A_3 &= P(a, x, g(f(y))) \\ A_4 &= P(x, y, g(f(b))) \\ A_5 &= P(a, f(x, a), g(z)) \\ A_6 &= P(y, h(y, f( \end{aligned}$$

$P_1$ : Au același simbol predicativ?

$P_2$ : Au aceeași aritate?

Cât timp  $\neq$ :

$$\left\{ \begin{aligned} \theta &= \{ \leftarrow \} \end{aligned} \right.$$

$$\left\{ \begin{aligned} \theta(A_1) &= \end{aligned} \right.$$

$$\left\{ \begin{aligned} \theta(A_2) &= \end{aligned} \right.$$

$A_1$  cu  $A_2$

$P_1$ : Da (P)

$P_2$ : Da (3)

$P_3$ :  $\left. \begin{array}{l} f\text{-simbol } f\text{-t.} \\ a\text{-constantă} \end{array} \right\} \Rightarrow A_1 \text{ și } A_2 \text{ nu sunt unificabile}$

$A_3$  cu  $A_4$

$P_1$ : Da (P)

$P_2$ : Da (3)

$$\mu(A_3) = P(a, a, g(f(y)))$$

$$\mu(A_4) = P(a, y, g(f(b)))$$

$$\mu' = [y \leftarrow a]$$

$$\mu'(\mu(A_3)) = P(a, a, g(f(a)))$$

$$\mu'(\mu(A_4)) = P(a, a, g(f(b)))$$

$a, b$  - ambele constante  
 $a \neq b$

$\Rightarrow A_3$  și  $A_4$  nu sunt unificabile

$A_5$  cu  $A_6$

$P_1$ : Da (P)

$P_2$ : Da (3)

$$\sigma_1(A_5) = P(a, h(x, u), g(z))$$

$$\sigma_1(A_6) = P(a, h(a, f(z)), g(x))$$

$$\alpha_2 = \{x \leftarrow \alpha\}$$

$$(\alpha_1(A_5)) = P(\alpha, h(\alpha, u), g(z))$$

$$\alpha_2(\alpha_1(A_6)) = P(\alpha, h(\alpha, f(z)), g(\alpha))$$